

Multi-Sensors Distributed Hypothesis Testing in the Low-Power Regime

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Abstract—We characterize the Stein-exponent of a distributed hypothesis testing scenario where two sensors transmit information through a memoryless multiple access channel (MAC) subject to a sublinear input cost constraint with respect to the number of channel uses and where the decision center has access to an additional local observation. Our main result provides conditions on the channel and cost functions for which the Stein-exponent of this distributed setup is no larger than the Stein-exponent of the local test at the decision center. The conditions are satisfied for additive noise MACs with generalized Gaussian noise under a p -th moment constraint (including the Gaussian channel with second-moment constraint) and for the class of fully-connected discrete memoryless multiple-access channels (DMMACs) (where all input pairs can induce all outputs) under arbitrary finite cost constraints. We further show that for not fully-connected DMMACs the Stein-exponent coincides with that of a setup with zero-rate noiseless communication links.

I. INTRODUCTION

Binary hypothesis testing aims to determine which of two joint distributions governs observed data. In this paper we study the performance of distributed binary hypothesis testing when sensors have stringent power budgets.

Binary hypothesis testing inherently encompasses two types of error: the type-I error probability which refers to the probability that the decision center declares $\hat{H} = 1$ while the true hypothesis underlying the observations is $H = 0$; and conversely. We are interested in asymmetric situations where type-II error probabilities are more harmful than type-I error probabilities, and thus consider the *Stein-exponent* [1], which measures the largest possible decay rate (in the number of observations n) of the type-II error probability under a fixed threshold $\epsilon \in [0, 1)$ on the type-I error probability.

The Stein-exponent is well-known for local tests where all the observations are locally available at the decision center [1], in which case it does not depend on the allowed type-I error threshold $\epsilon \in [0, 1)$. For most distributed scenarios where part of the observations are located at remote sensors and need to be communicated to the decision center, the Stein-exponent is however still unknown. Notable exceptions are, for example, [2]–[9], which make different assumptions on the communication from the sensors to the decision center.

A line of work that is closely related to the present work studied the Stein-exponent in a noiseless-link setup, but under

the constraint that the number of bits communicated from the sensor to the decision center grows only sublinearly in n . This setting is commonly known as *noiseless zero-rate communication*, and [2], [4] characterized the Stein-exponent for a broad class of source distributions.

In this work we will relax the assumption of noise-free communication. In the information-theoretic literature, noisy communication channels for hypothesis testing have been explored in [5], [10], presenting general lower bounds and exact characterizations of the Stein-exponent for special source distributions. However, in these previous works, the number of channel uses is linear in the observation length and no cost constraints are imposed.

A recent work [11] also focuses on a single-sensor setup where communication is over a discrete-memoryless channel (DMC). It is shown that when the sensor can only send a sublinear number of inputs or the inputs are subject to a sublinear cost constraint, the Stein-exponent only depends on whether the DMC is fully-connected, i.e., each input symbol induces each output symbol with positive probability, or not. For fully-connected DMCs, the Stein-exponent completely degrades to the Stein-exponent of the local test at the decision center rendering the sensor and its communication useless.

In this paper, we extend the results in [11] to scenarios with multiple sensors that can communicate to the decision center over a continuous or discrete memoryless MAC. For a large class of MACs, we show a complete degradation of the Stein-exponent to the one of a local test, rendering the communication and the sensors useless. As we further show, this is in particular the case for

- continuous-valued MACs with p -th order generalized Gaussian noise subject to a p -th moment cost constraints;
- fully-connected DMMACs with arbitrary cost constraints.

For other connectivity patterns of the DMMAC, the Stein-exponent over DMMACs coincides with the Stein-exponent of the communication scenario where either one or both sensors can communicate to the decision center over zero-rate noiseless links, as in [2].

The rest of this paper is organized as follows. Section II introduces notation, Section III describes the general problem setup, and Section IV presents our results.

II. NOTATION

Random variables are denoted by uppercase letters like U , while their realizations are denoted by lowercase letters like u . We abbreviate $(u_1, \dots, u_n)$ by u^n and $(U_1, \dots, U_n)$ by U^n . The set of real and nonnegative real numbers are denoted $\mathbb{R}$ and $\mathbb{R}^+$. We use P_U to denote the distribution of the random variable U and the associated Lebesgue-Stieltjes measure [12, Section 3.11] dP_U . When it exists, the corresponding probability density function is denoted p_U . The notation $U \sim P_U^{\otimes n}$ indicates that the variables U^n are *independent and identically distributed*, abbreviated as *i.i.d.*, according to P_U . Let P_1 and P_2 be two distributions over $\mathcal{U}$. Suppose that P_1 is absolutely continuous with respect to P_2 (denoted $P_1 \ll P_2$), then the Kullback-Leibler divergence [13], [14] between P_1 and P_2 is defined as

$$D(P_1 \| P_2) = \int_{\mathcal{U}} \ln \left(\frac{dP_1}{dP_2} \right) dP_1, \quad (1)$$

where $\frac{dP_1}{dP_2}$ is the Radon–Nikodym derivative [15, p. 128] of P_1 with respect to P_2 . We denote the quasi p -norms by $\|a^n\|_p = (\sum_{i=1}^n |a_i|^p)^{1/p}$ for any $a^n \in \mathbb{R}^n$.

III. GENERAL MODEL

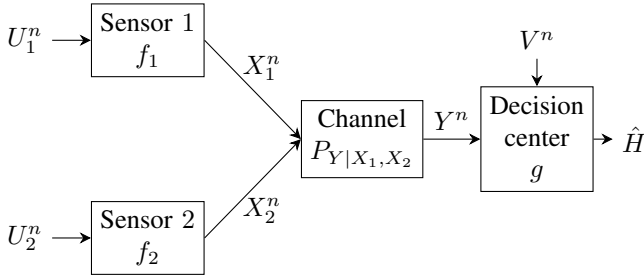


Fig. 1. Distributed hypothesis testing over a memoryless channel.

We consider the setup of Figure 1 where two sensors observe random sequences U_1^n and U_2^n , respectively, and a decision center observes the random sequence V^n . The observations take value in finite sets $\mathcal{U}_1^n, \mathcal{U}_2^n$, and $\mathcal{V}^n$, respectively, and their distribution depends on a binary hypothesis $H \in \{0, 1\}$ in the sense that:

$$\text{under } H = 0: (U_1^n, U_2^n, V^n) \sim P_{U_1, U_2, V}^{\otimes n} \quad (2a)$$

$$\text{under } H = 1: (U_1^n, U_2^n, V^n) \sim Q_{U_1, U_2, V}^{\otimes n}, \quad (2b)$$

where we assume that $P_{U_1, U_2, V} \ll Q_{U_1, U_2, V}$.

The two sensors communicate over a memoryless multiple-access channel (MAC) with a decision center, which then attempts to guess the hypothesis H based on its own local observations and on its observed channel outputs. More formally, each Sensor $\ell \in \{1, 2\}$ produces a sequence of channel inputs $X_\ell^n = f_\ell(U_\ell^n)$ using an encoding function of the form

$$f_\ell: \mathcal{U}_\ell^n \rightarrow \mathcal{X}_\ell^n$$

$$u_\ell^n \mapsto x_\ell^n = f_\ell(u_\ell^n), \quad \ell \in \{1, 2\}. \quad (3)$$

The input sequences are subject to stringent cost constraints. That means, we are given per-symbol input cost functions

$$c_\ell: \mathcal{X}_\ell \rightarrow \mathbb{R}^+, \quad (4)$$

where we assume that there exists a unique zero-symbol $x_\ell \in \mathcal{X}_\ell$ with $c_\ell(x_\ell) = 0$. For simplicity, we call 0 the zero-symbol for both c_1 and c_2 . We are also given stringent cost budgets $\Gamma_1(n) > 0$ and $\Gamma_2(n)$ that grow sublinearly in n :

$$\lim_{n \rightarrow \infty} \Gamma_\ell(n) = \infty \text{ and } \lim_{n \rightarrow \infty} \frac{\Gamma_\ell(n)}{n} = 0, \quad \ell \in \{1, 2\}. \quad (5)$$

For each u_ℓ^n , the cost constraints impose that $f_\ell(u_\ell^n)$ lies in

$$\tilde{\mathcal{X}}_\ell^n \triangleq \left\{ x_\ell^n \in \mathcal{X}_\ell^n: \sum_{i=1}^n c_\ell(x_{\ell,i}) \leq \Gamma_\ell(n) \right\}, \quad \ell \in \{1, 2\}. \quad (6)$$

Given channel inputs x_1^n and x_2^n , the decision center observes a random output sequence Y^n with each Y_i distributed according to the conditional distribution $P_{Y|X_1, X_2}(\cdot | x_1, x_2)$.

The decision center applies a guessing function of the form

$$g: \mathcal{Y}^n \times \mathcal{V}^n \rightarrow \{0, 1\} \quad (7)$$

to produce a guess $\hat{H} = g(Y^n, V^n)$. The decision's type-I (false positive) and type-II (false negative) error probabilities are respectively denoted

$$\alpha_n = \mathbb{P}[g(Y^n, V^n) = 1 \mid H = 0] \quad (8a)$$

$$\beta_n = \mathbb{P}[g(Y^n, V^n) = 0 \mid H = 1]. \quad (8b)$$

Definition 1 Given $\epsilon \in [0, 1)$, a type-II error exponent $\theta > 0$ is called $(\epsilon, \Gamma_1(\cdot), \Gamma_2(\cdot))$ -achievable if there exists a sequence (in the blocklength n) of encoding and decision functions (f_1, f_2, g) satisfying the sublinear cost constraint (6) and

$$\lim_{n \rightarrow \infty} \overline{\alpha_n} \leq \epsilon \quad (9a)$$

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \ln \beta_n \geq \theta. \quad (9b)$$

The supremum over all $(\epsilon, \Gamma_1(\cdot), \Gamma_2(\cdot))$ -achievable type-II error exponents θ is denoted $\theta_{\text{sublin}}^*(\epsilon, \Gamma_1(\cdot), \Gamma_2(\cdot))$.

We shall see that $\theta_{\text{sublin}}^*(\epsilon, \Gamma_1(\cdot), \Gamma_2(\cdot))$ is the same for all functions $\Gamma_1(\cdot)$ and $\Gamma_2(\cdot)$ satisfying (5) and all $\epsilon \in [0, 1)$. We therefore simply write θ_{sublin}^* .

IV. RESULTS

We introduce two preliminary propositions.

Proposition 1 Allowing the decision center to take a randomized decision does not increase the Stein-exponent θ_{sublin}^* .

Proof: See Appendix A in the extended version [16]. ■

We have the following general result on the type-II error exponent showing that for certain channels $P_{Y|X_1, X_2}$, communication from the sensors does not increase the Stein-exponent due to the stringent cost constraints (5).

Proposition 2 Assume that there exists a sequence of sets $\mathcal{D}_n \subseteq \mathcal{Y}^n$ satisfying

$$\lim_{n \rightarrow \infty} \mathbb{P}[Y^n \notin \mathcal{D}_n] = 0 \quad (10)$$

and for all input sequences $x_1^n, \tilde{x}_1^n \in \tilde{\mathcal{X}}_1^n$ and $x_2^n, \tilde{x}_2^n \in \tilde{\mathcal{X}}_2^n$:

$$\overline{\lim}_{n \rightarrow \infty} -\frac{1}{n} \ln \left(\frac{dP_{Y^n|X_1^n, X_2^n}(y^n|x_1^n, x_2^n)}{dP_{Y^n|X_1^n, X_2^n}(y^n|\tilde{x}_1^n, \tilde{x}_2^n)} \right) \leq 0, \quad \forall y^n \in \mathcal{D}_n. \quad (11)$$

Then, for any $\epsilon \in [0, 1)$, $\Gamma_1(\cdot)$ and $\Gamma_2(\cdot)$ satisfying (5), the optimal type-II error exponent is

$$\theta_{\text{sublin}}^* = D(P_V \| Q_V). \quad (12)$$

Proof: See Appendix A. ■

A. Generalized Gaussian channels with p -th moment costs

Consider the MAC

$$Y_i = h_1 x_{1,i} + h_2 x_{2,i} + Z_i, \quad i \in \{1, \dots, n\}, \quad (13)$$

where h_1 and h_2 are given non-zero real channel coefficients, $(Z_i)_{i \geq 1}$ is an i.i.d. noise sequence independent of the inputs x_1^n and x_2^n , and each Z_i follows a generalized Gaussian distribution [17], [18] with parameters $p, \sigma > 0$, i.e., of probability density

$$p_Z(z) = \frac{c_p}{\sigma} e^{-\frac{|z|^p}{2\sigma^p}}, \quad z \in \mathbb{R}, \quad (14)$$

where $c_p \triangleq p \left(2^{\frac{p+1}{p}} \Gamma\left(\frac{1}{p}\right) \right)^{-1}$ and Γ denotes the gamma function [19]. A p -th moment cost constraint is imposed on the input sequences:

$$\|x_\ell^n\|_p^p \leq \Gamma_\ell(n), \quad \ell \in \{1, 2\}, \quad (15)$$

i.e., $c_\ell(x) = |x|^p, \ell = 1, 2$.

For $p = 2$, the noise is Gaussian and the cost constraint is a standard average block-power constraint.

Theorem 1 For the above generalized Gaussian setup, for any $p > 0$, $\epsilon \in [0, 1)$, and p -th moment cost constraints $\Gamma_1(\cdot)$ and $\Gamma_2(\cdot)$ satisfying (5):

$$\theta_{\text{sublin}}^* = D(P_V \| Q_V). \quad (16)$$

Proof: The detailed proof is available in the extended version of this paper [16, Section VI]. Omitting long computations, a sketch of the proof is given in Appendix B. ■

B. Discrete memoryless channels with finite costs

Input and output sets $\mathcal{X}_1, \mathcal{X}_2$, and $\mathcal{Y}$ are finite and the channel is described by the transition law $P_{Y|X_1, X_2}$. Cost functions are bounded $c_\ell(x_\ell) < \infty$ for all x_ℓ and $\ell = 1, 2$. As we shall see, θ_{sublin}^* depends on the connectivity of the DMMAC. We therefore define the following classes of channels.

- The set $\mathcal{C}_{\text{full}}$ contains all DMMACs $P_{Y|X_1, X_2}$ for which

$$P_{Y|X_1, X_2}(y|x_1, x_2) > 0, \forall y, x_1, x_2 \in \mathcal{Y} \times \mathcal{X}_1 \times \mathcal{X}_2. \quad (17)$$

- The set $\mathcal{C}_{\text{sparse}}$ contains all DMMACs $P_{Y|X_1, X_2}$ where there exists (not necessarily distinct) $x_1, x'_1, \tilde{x}_1 \in \mathcal{X}_1$, $x_2, x'_2, \tilde{x}_2 \in \mathcal{X}_2$ and $y, \tilde{y} \in \mathcal{Y}$ such that

$$P_{Y|X_1, X_2}(y|x_1, \tilde{x}_2) = 0 \quad (18a)$$

$$P_{Y|X_1, X_2}(y|x'_1, \tilde{x}_2) > 0 \quad (18b)$$

$$P_{Y|X_1, X_2}(\tilde{y}|\tilde{x}_1, x_2) = 0 \quad (18c)$$

$$P_{Y|X_1, X_2}(\tilde{y}|\tilde{x}_1, x'_2) > 0. \quad (18d)$$

- The set $\mathcal{C}_{\text{sparse-full}}$ contains all DMMACs $P_{Y|X_1, X_2}$ for which there exists $x_1, x'_1 \in \mathcal{X}_1$, $\tilde{x}_2 \in \mathcal{X}_2$, $y^* \in \mathcal{Y}$ such that

$$P_{Y|X_1, X_2}(y^*|x_1, \tilde{x}_2) = 0 \quad (19a)$$

$$P_{Y|X_1, X_2}(y^*|x'_1, \tilde{x}_2) > 0, \quad (19b)$$

and for any pair $(x_1, y) \in \mathcal{X}_1 \times \mathcal{Y}$ we either have

$$P_{Y|X_1, X_2}(y|x_1, x_2) = 0, \quad \forall x_2 \in \mathcal{X}_2 \quad (20a)$$

or

$$P_{Y|X_1, X_2}(y|x_1, x_2) > 0, \quad \forall x_2 \in \mathcal{X}_2. \quad (20b)$$

- Finally, the set $\mathcal{C}_{\text{full-sparse}}$ contains all DMMACs $P_{Y|X_1, X_2}$ for which there exists $\tilde{x}_1 \in \mathcal{X}_1$, $x_2, x'_2 \in \mathcal{X}_2$ and $y^* \in \mathcal{Y}$ such that

$$P_{Y|X_1, X_2}(y^*|\tilde{x}_1, x_2) = 0 \quad (21a)$$

$$P_{Y|X_1, X_2}(y^*|\tilde{x}_1, x'_2) > 0, \quad (21b)$$

and for any pair $(x_2, y) \in \mathcal{X}_2 \times \mathcal{Y}$ we either have

$$P_{Y|X_1, X_2}(y|x_1, x_2) = 0, \quad \forall x_1 \in \mathcal{X}_1 \quad (22a)$$

or

$$P_{Y|X_1, X_2}(y|x_1, x_2) > 0, \quad \forall x_1 \in \mathcal{X}_1. \quad (22b)$$

To see that the four sets $\mathcal{C}_{\text{full}}, \mathcal{C}_{\text{sparse}}, \mathcal{C}_{\text{sparse-full}}, \mathcal{C}_{\text{full-sparse}}$ are a partition of all channel laws, notice that for fixed y^* and $\tilde{x}_2$,

$$\begin{aligned} \{ \exists x_1, \tilde{x}_1: & P_{Y|X_1, X_2}(y^*|x_1, \tilde{x}_2) = 0 \\ & \text{and } P_{Y|X_1, X_2}(y^*|x'_1, \tilde{x}_2) > 0 \} \end{aligned} \quad (23)$$

is the complementary event of

$$\begin{aligned} \{ \forall x_1: & P_{Y|X_1, X_2}(y^*|x_1, \tilde{x}_2) = 0 \} \\ \cup & \{ \forall x_1: P_{Y|X_1, X_2}(y^*|x_1, \tilde{x}_2) > 0 \}. \end{aligned} \quad (24)$$

Theorem 2 1) If $P_{Y|X_1, X_2} \in \mathcal{C}_{\text{full}}$, then

$$\theta_{\text{sublin}}^* = D(P_V \| Q_V). \quad (25)$$

2) If $P_{Y|X_1, X_2} \in \mathcal{C}_{\text{sparse}}$, then

$$\theta_{\text{sublin}}^* = \min D(\tilde{P}_{U_1, U_2, V} \| Q_{U_1, U_2, V}), \quad (26)$$

where the minimum is taken over all distributions $\tilde{P}_{U_1, U_2, V}$ satisfying $\tilde{P}_{U_1} = P_{U_1}$, $\tilde{P}_{U_2} = P_{U_2}$, $\tilde{P}_V = P_V$. Interestingly, this coincides with the Stein-exponent in the noise-less zero-rate communication setup [2].

3) If $P_{Y|X_1, X_2} \in \mathcal{C}_{\text{sparse-full}}$, then

$$\theta_{\text{sublin}}^* = \min D\left(\tilde{P}_{V, U_1} \| Q_{V, U_1}\right), \quad (27)$$

where the minimum is taken over all distributions $\tilde{P}_{V, U_1}$ satisfying $\tilde{P}_{U_1} = P_{U_1}$, $\tilde{P}_V = P_V$.

4) If $P_{Y|X_1, X_2} \in \mathcal{C}_{\text{full-sparse}}$, then

$$\theta_{\text{sublin}}^* = \min D\left(\tilde{P}_{V, U_2} \| Q_{V, U_2}\right), \quad (28)$$

where the minimum is taken over all distributions $\tilde{P}_{V, U_2}$ satisfying $\tilde{P}_{U_2} = P_{U_2}$, $\tilde{P}_V = P_V$.

Proof: We only provide a sketch of the proof here. For detailed proofs see the extended version [16, Section VII].

The converse for 1) can be shown by applying Proposition 2 for $\mathcal{D}_n = \mathcal{Y}_n$. Achievability is proved by a local test at the decision center.

For 2) the converse is proved by giving the decision center direct access to the sensor's inputs, and showing that the stringent cost constraints imply that $|\mathcal{X}_\ell|$ are sublinear in n , reducing the setup to the zero-rate scenario in [2]. Achievability can be shown using a scheme, where each sensor performs a local test based on its observations, and communicates the outcome to the decision during the first (Sensor 1) or second (Sensor 2) block of $k = 1/2 \cdot \min_{\ell=1,2} \frac{\Gamma_\ell}{\max_x c_\ell(x)}$ channel uses. Let x_ℓ, x'_ℓ , and $\tilde{x}_\ell$ satisfy (18). To indicate $H = 0$ for its local test, Sensor ℓ sends the x'_ℓ sequence and otherwise x_ℓ^k . In the block dedicated to the other sensor it sends $\tilde{x}_\ell^k$. By assumptions (18), communication is error-free whenever the sensors' local tests suggest $H = 1$.

For 3), achievability can be proved by applying above scheme only during the first $k/2$ channel uses. The converse is proved by providing the decision center direct access to Sensor 1's input X_1^n , and then relating type-I and type-II error probabilities in this new setup to the error probabilities in a setup where instead of outputs Y^n , the decision center observes $\tilde{Y}^n$ which under both distributions follows the same law (in fact the law of Y^n under $H = 0$ in the original setup). Similarly as in Appendix 2, $\tilde{Y}^n$ can be considered a local randomness, and thus does not increase the Stein-exponent.

The proof of 4) is symmetric to the proof of 3) exchanging the roles of Sensors 1 and 2. ■

Example 1 Consider the DMMAC

$$Y = S_1 \cdot x_1 + S_2 \cdot x_2 + Z, \quad (29)$$

where multiplications and additions are the standard operations in $\mathbb{R}$ and Z is a discrete noise over any discrete and finite set $\mathcal{Z} \subset \mathbb{R}$. Inputs x_1 and x_2 as well as the auxiliary variables S_1 and S_2 take value in $\{-1, 1\}$. Assuming that Z is not deterministic, depending on the law of the "states" S_1 and S_2 , the DMMAC belongs to one of the four classes above:

- For S_1 and S_2 both deterministic, the DMMAC belongs to $\mathcal{C}_{\text{sparse}}$.
- For S_1 and S_2 both non-deterministic, the DMMAC belongs to $\mathcal{C}_{\text{full}}$.

- For S_1 deterministic and S_2 non-deterministic, the DM-MAC belongs to $\mathcal{C}_{\text{sparse-full}}$.
- For S_1 non-deterministic and S_2 deterministic, the DM-MAC belongs to $\mathcal{C}_{\text{full-sparse}}$.

APPENDIX

A. Proof of Proposition 2

Achievability follows directly from Stein's lemma, where the decision center can ignore the channel outputs Y^n . The converse is proved by relating the type-I and type-II error probabilities of our distributed hypothesis testing problem to the error probabilities of a randomized local hypothesis testing setup.

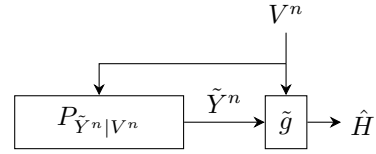


Fig. 2. Randomized local hypothesis test.

We start our proof by fixing any sequence of encoding and decision functions $\{f_1, f_2, g\}_{n=1}^\infty$ with type-I error satisfying $\lim_{n \rightarrow \infty} \alpha_n \leq \epsilon$. For each observation $v^n \in \mathcal{V}^n$, define the acceptance regions

$$\mathcal{A}(v^n) \triangleq \{y^n \in \mathbb{R}^n : g(v^n, y^n) = 0\}. \quad (30)$$

Let $P_{\tilde{Y}^n|V^n} \triangleq P_{Y^n|V^n, H}(\cdot, 0)$ denote the joint distribution induced on $U^n, V^n, X_1^n, X_2^n, Y^n$ by the chosen encodings f_1, f_2 under $H = 0$. Introduce then the new binary hypothesis testing setup depicted in Figure 2 without any sensor. As before, under $H = 0$ the decision center observes V^n i.i.d. $\sim P_V$ and under $H = 1$ it observes V^n i.i.d. $\sim Q_V$. Moreover, under both hypotheses, it has access to an additional local randomness $\tilde{Y}^n$ that is obtained from V^n based on the non-i.i.d. conditional distribution $P_{\tilde{Y}^n|V^n}$, irrespectively of the hypothesis H . The decision function in this auxiliary setup is of the form $\tilde{g}: \mathcal{V}^n \times \mathcal{Y}^n \rightarrow \{0, 1\}$ and we denote its type-I and type-II error probabilities by $\tilde{\alpha}_n$ and $\tilde{\beta}_n$:

$$\tilde{\alpha}_n \triangleq \mathbb{P}[\tilde{g}(V^n, \tilde{Y}^n) = 1 | H = 0] \quad (31a)$$

$$\tilde{\beta}_n \triangleq \mathbb{P}[\tilde{g}(V^n, \tilde{Y}^n) = 0 | H = 1]. \quad (31b)$$

Choosing the decision rule

$$\tilde{g}(v^n, y^n) = g(v^n, y^n) \cdot \mathbb{1}\{y^n \in \mathcal{D}_{\delta, n}\}, \quad (32)$$

we have:

$$\begin{aligned} 1 - \alpha_n &\leq \mathbb{P}[g(V^n, Y^n) = 0, Y^n \in \mathcal{D}_n | H = 0] \\ &\quad + \mathbb{P}[Y^n \notin \mathcal{D}_n | H = 0] \\ &= \sum_{v^n \in \mathcal{V}^n} \left(\mathbb{P}[g(V^n, \tilde{Y}^n) = 0, \tilde{Y}^n \in \mathcal{D}_n | V^n = v^n, H = 0] \right) \end{aligned} \quad (33)$$

$$\cdot P_V^{\otimes n}(v^n)) + \mathbb{P}[Y^n \notin \mathcal{D}_n \mid H = 0] \quad (34)$$

$$= 1 - \tilde{\alpha}_n + \mathbb{P}[Y^n \notin \mathcal{D}_n \mid H = 0], \quad (35)$$

where (34) holds because we defined $\tilde{Y}^n$ to have the same conditional distribution given V^n as Y^n under hypothesis $H = 0$ and (35) holds by the definition of the $\tilde{g}$ function in (32). Combining above inequality with (10), we deduce that

$$\lim_{n \rightarrow \infty} \tilde{\alpha}_n \leq \lim_{n \rightarrow \infty} \alpha_n \leq \epsilon. \quad (36)$$

Define

$$\zeta \triangleq \inf_{\substack{y^n \in \mathcal{D}_n, \\ x_1^n, \tilde{x}_1^n \in \tilde{\mathcal{X}}_1^n, \\ x_2^n, \tilde{x}_2^n \in \tilde{\mathcal{X}}_2^n}} \frac{dP_{Y^n|X_1^n, X_2^n}(y^n|x_1^n, x_2^n)}{dP_{Y^n|X_1^n, X_2^n}(y^n|\tilde{x}_1^n, \tilde{x}_2^n)}. \quad (37)$$

We can now bound β_n in terms of $\tilde{\beta}_n$:

$$\beta_n \geq \sum_{v^n \in \mathcal{V}^n} \left(Q_V^{\otimes n}(v^n) \int_{\mathcal{A}(v^n) \cap \mathcal{D}_n} dP_{Y^n|V^n, H}(y^n|v^n, 1) \right) \quad (38)$$

$$\geq \zeta \sum_{v^n \in \mathcal{V}^n} \left(Q_V^{\otimes n}(v^n) \int_{\mathcal{A}(v^n) \cap \mathcal{D}_n} \sum_{(x_1^n, x_2^n) \in \tilde{\mathcal{X}}_1^n \times \tilde{\mathcal{X}}_2^n} (P_{X_1^n, X_2^n|V^n, H}(x_1^n, x_2^n|v^n, 0) \cdot dP_{Y^n|X_1^n, X_2^n}(y^n|x_1^n, x_2^n)) \right) \quad (39)$$

$$= \zeta \cdot \tilde{\beta}_n, \quad (40)$$

where (38) holds by restricting the integral; (39) by the definition of ζ in (37) and because for any bounded function f the expectations of f with respect to any two measures μ and ν satisfy $\mathbb{E}_\nu[f]/\mathbb{E}_\mu[f] \geq f_{\min}/f_{\max}$ where $f_{\min}$ and $f_{\max}$ denote the infimum and supremum of f ; (40) by the definition of $\tilde{\beta}_n$.

Notice now that we can specialize Proposition 1 to a setup where the decision center can base its decision only on the local observation V^n and the local randomness $\tilde{Y}^n$. We conclude that $\frac{1}{n} \ln \tilde{\beta}_n$ is asymptotically upper bounded by the Stein-exponent of a non-random test based on V^n only, i.e.,

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \ln \tilde{\beta}_n \leq D(P_V \| Q_V). \quad (41)$$

In addition, Assumption (11) ensures that

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \ln \zeta \leq 0. \quad (42)$$

Plugging (41) and (42) into (40) concludes the converse proof.

B. Proof of Theorem 1

Let $x_1^n, \tilde{x}_1^n \in \tilde{\mathcal{X}}_1^n$ and $x_2^n, \tilde{x}_2^n \in \tilde{\mathcal{X}}_2^n$. Then, defining $b_i \triangleq h_1 x_{1,i} + h_2 x_{2,i}$ and $\tilde{b}_i \triangleq h_1 \tilde{x}_{1,i} + h_2 \tilde{x}_{2,i}$, for $i \in \{1, \dots, n\}$, as well as $b^n = h_1 x_1^n + h_2 x_2^n$ and $\tilde{b}^n = h_1 \tilde{x}_1^n + h_2 \tilde{x}_2^n$, we notice that the input power constraints (15) imply:

$$\max\{\|b^n\|_p^p, \|\tilde{b}^n\|_p^p\} \leq 2^p(h_1^p \Gamma_1(n) + h_2^p \Gamma_2(n)). \quad (43)$$

We will use this inequality in the following converse steps.

Case $p \in (0, 1]$: The converse follows directly from Proposition 2 by choosing $\mathcal{D}_n = \mathcal{Y}^n$. In fact, for any input sequences $x_1^n, \tilde{x}_1^n \in \tilde{\mathcal{X}}_1^n$ and $x_2^n, \tilde{x}_2^n \in \tilde{\mathcal{X}}_2^n$ and any output sequence $y^n \in \mathbb{R}^n$:

$$\frac{p_{Y|X_1, X_2}^{\otimes n}(y^n|x_1^n, x_2^n)}{p_{Y|X_1, X_2}^{\otimes n}(y^n|\tilde{x}_1^n, \tilde{x}_2^n)} \geq \exp\left(-\frac{\|b^n - \tilde{b}^n\|_p^p}{2\sigma^p}\right) \quad (44)$$

$$\geq \exp\left(-\frac{\|b^n\|_p^p + \|\tilde{b}^n\|_p^p}{2\sigma^p}\right) \quad (45)$$

$$\geq \exp\left(-\frac{2^p(h_1^p \Gamma_1(n) + h_2^p \Gamma_2(n))}{\sigma^p}\right). \quad (46)$$

Here, (46) holds by (43).

The proof is concluded by noting that (46) implies (11) because p, h_1, h_2, σ are fixed and by our assumption (5).

Case $p > 1$: Follows from Proposition 2 by choosing

$$\mathcal{D}_n = \{y^n \in \mathbb{R}^n : \|y^n\|_p^p \leq \nu\}, \quad (47)$$

where

$$\nu \triangleq 2^{2p-2}(h_1^p \Gamma_1(n) + h_2^p \Gamma_2(n)) + 2^{p-1}\left(n \frac{2\sigma^p}{p} + \delta n\right) \quad (48)$$

for an arbitrary small number $\delta > 0$.

We check that the proposed set $\mathcal{D}_n$ satisfies the two Conditions (10) and (11) in Proposition 2. As shown in detail in [16], for any $x_1^n, \tilde{x}_1^n \in \tilde{\mathcal{X}}_1^n$ and $x_2^n, \tilde{x}_2^n \in \tilde{\mathcal{X}}_2^n$ one can show:

$$\begin{aligned} & \frac{p_{Y|X}^{\otimes n}(y^n|x^n)}{p_{Y|X}^{\otimes n}(y^n|\tilde{x}^n)} \\ & \geq \exp\left(-\frac{2^{p-2}p}{\sigma^p}\left(4 \cdot 2^p(h_1^p \Gamma_1(n) + h_2^p \Gamma_2(n)) \right. \right. \\ & \quad \left. \left. + 2(h_1^p \Gamma_1(n) + h_2^p \Gamma_2(n))^{\frac{1}{p}} \nu^{\frac{p-1}{p}}\right)\right). \end{aligned} \quad (49)$$

Recalling the choice of ν in (48) and the stringent power constraints (5), we can conclude from (49) that Condition (11) is satisfied. The remaining Condition (10) holds because

$$\mathbb{P}[Y^n \notin \mathcal{D}_n \mid H = 0] \leq \mathbb{P}\left[\frac{\sum_{i=1}^n (|Z_i|^p - \mathbb{E}[|Z_i|^p])}{n} \geq \delta\right], \quad (50)$$

where the right-hand side vanishes as $n \rightarrow \infty$ by the weak law of large numbers.

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